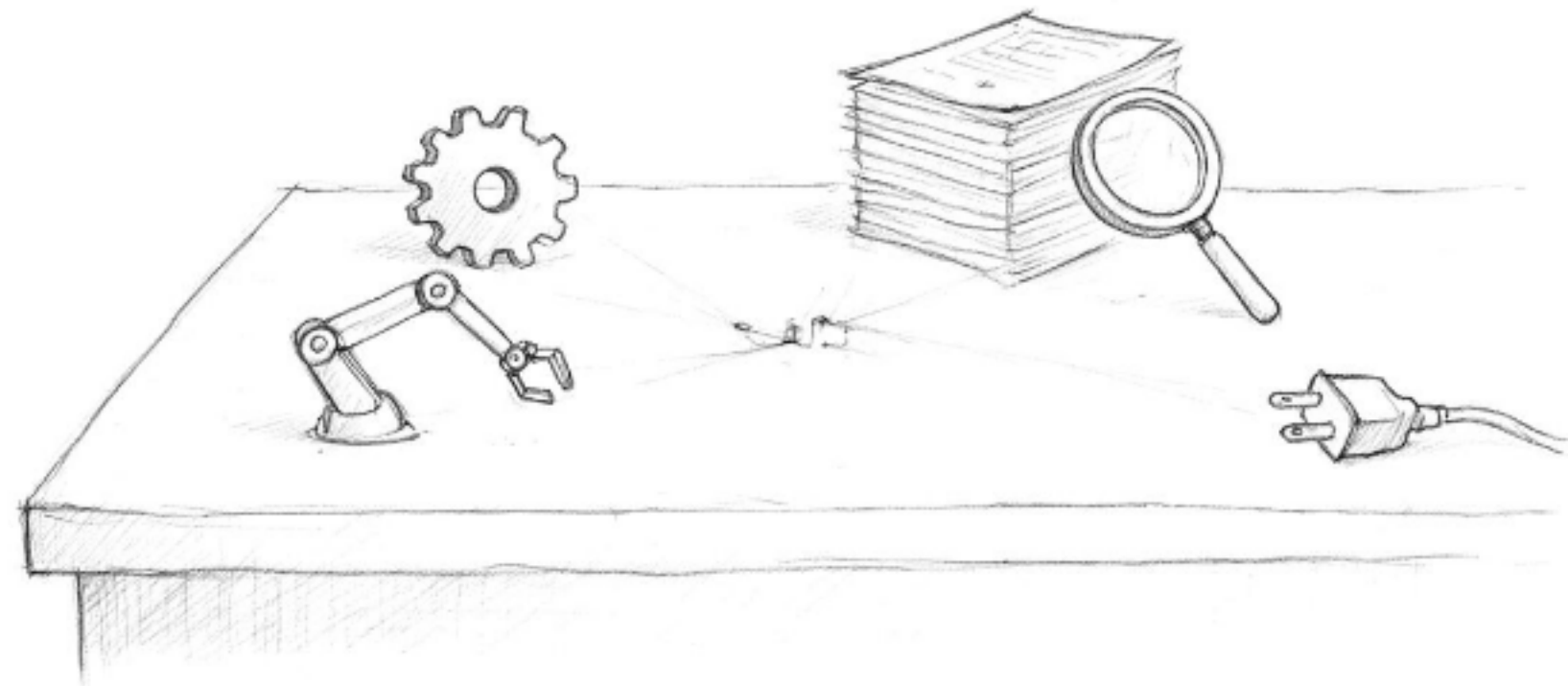
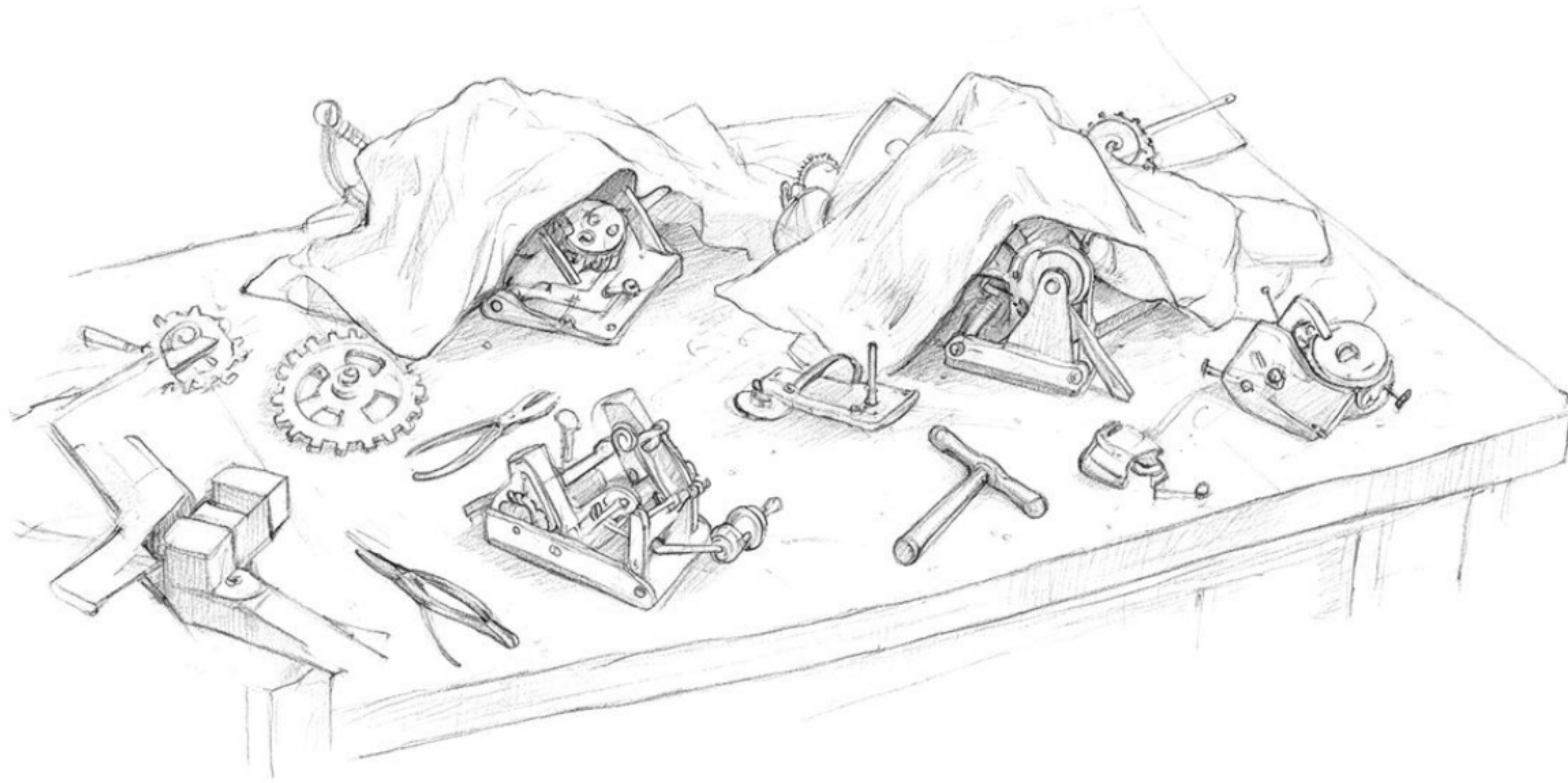


Capstone — Your AI Assistant, End to End



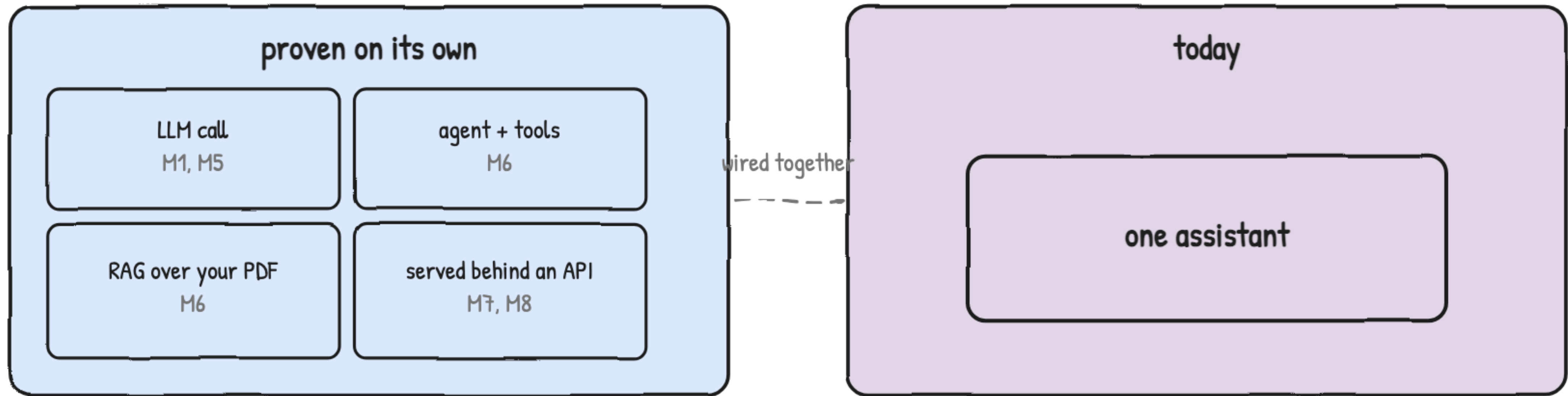
Four real labs. One assistant, wired together.

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What happens when five separate things become one thing?

Four real things, each one worked, each one torn down

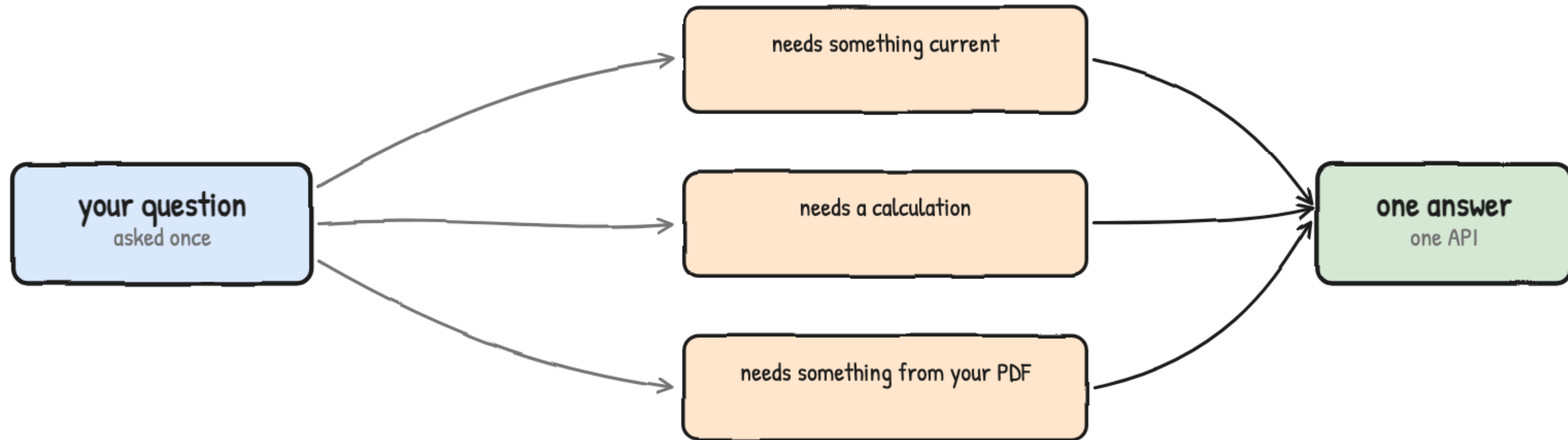


Proven on its own. Never wired together, until today.

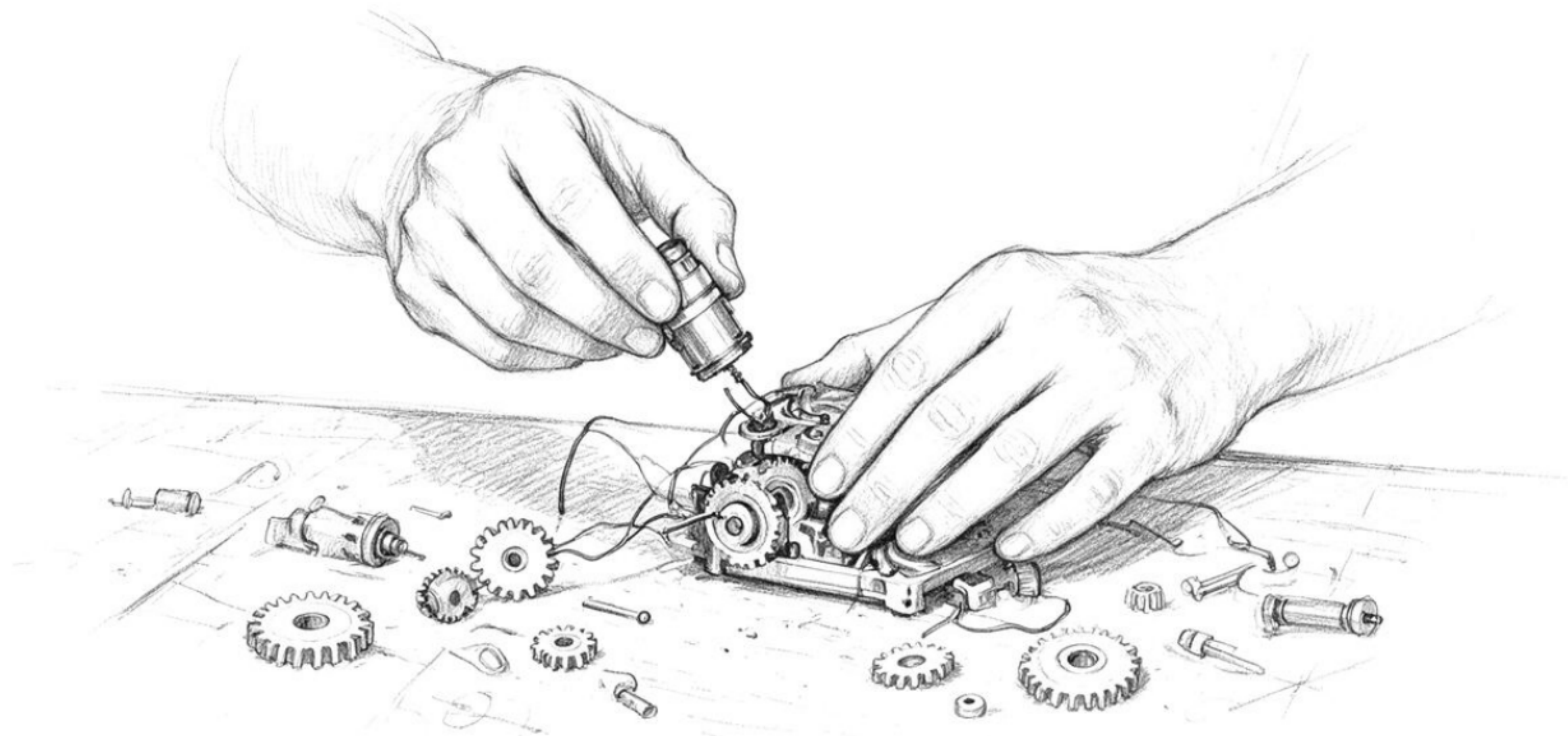


The capstone brief: what you're building

One assistant, three kinds of questions

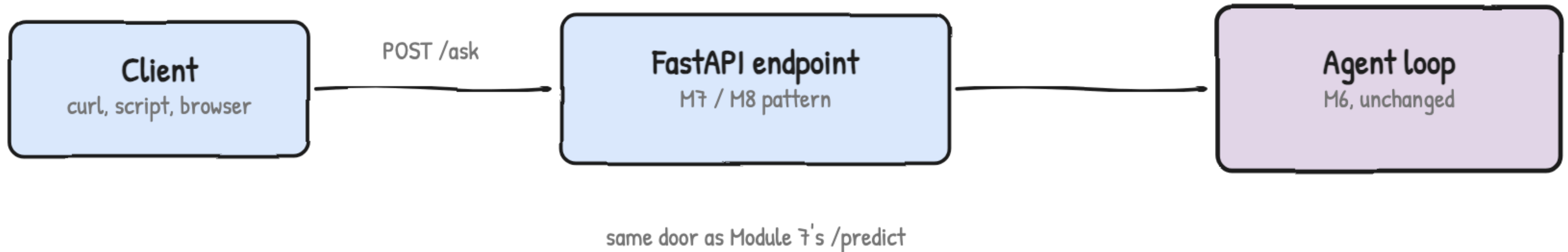


One door in, the same way every time.



The architecture, end to end

Every request needs one door in



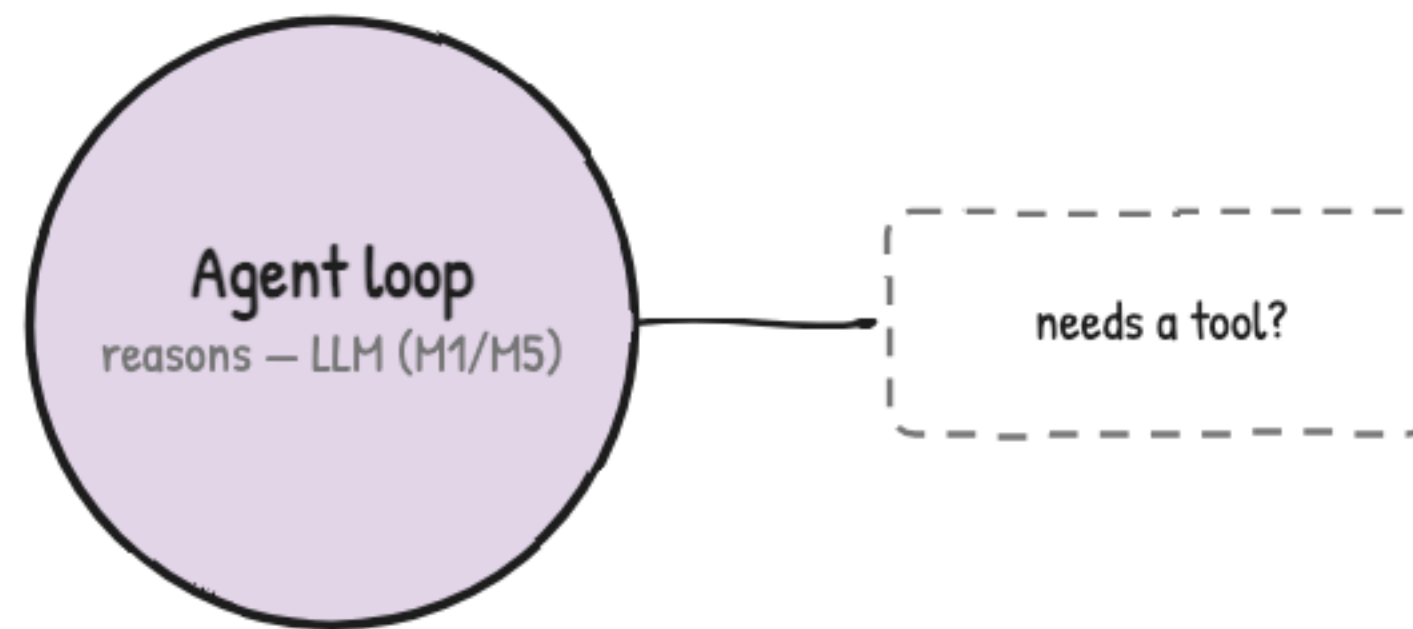
Same door Module 7 built for /predict.

Six modules converge behind one loop



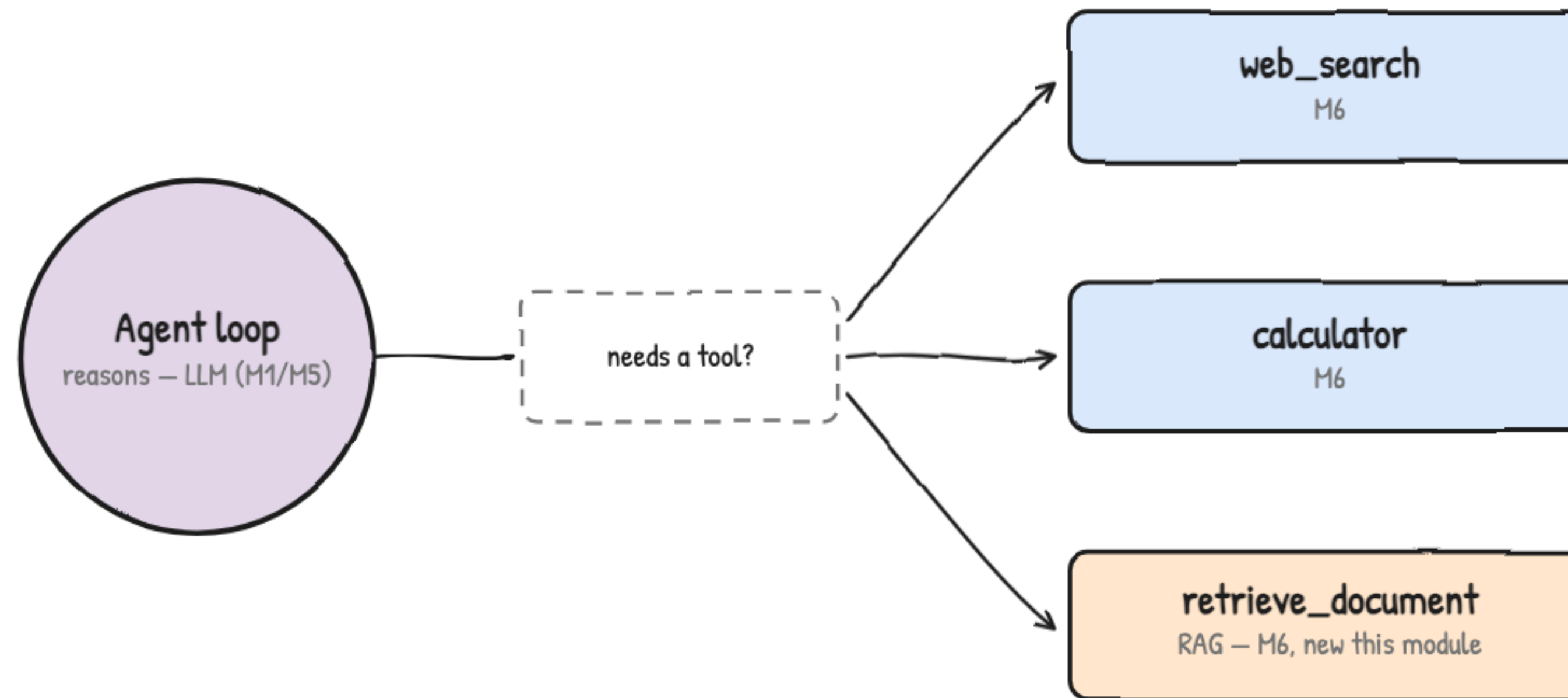
Read it left to right, one hop at a time.

Six modules converge behind one loop



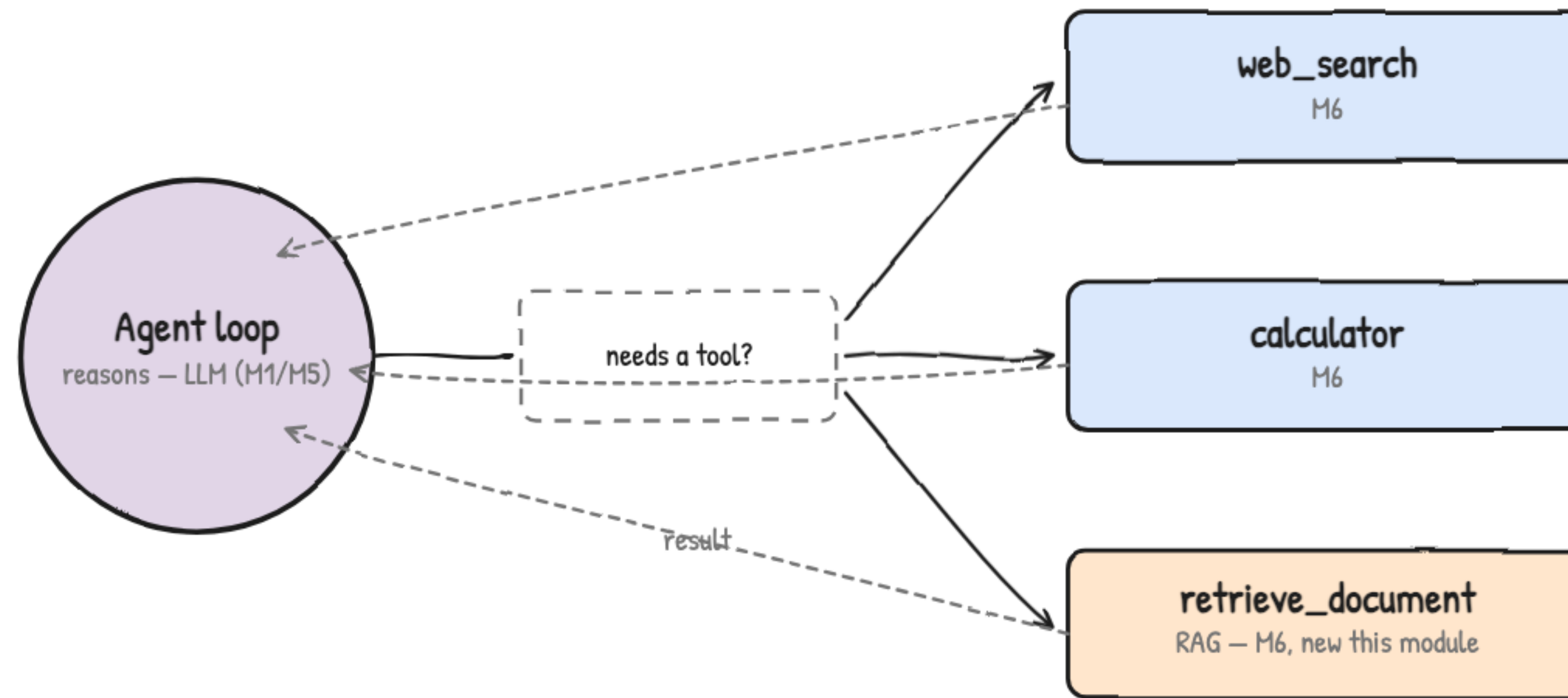
Read it left to right, one hop at a time.

Six modules converge behind one loop



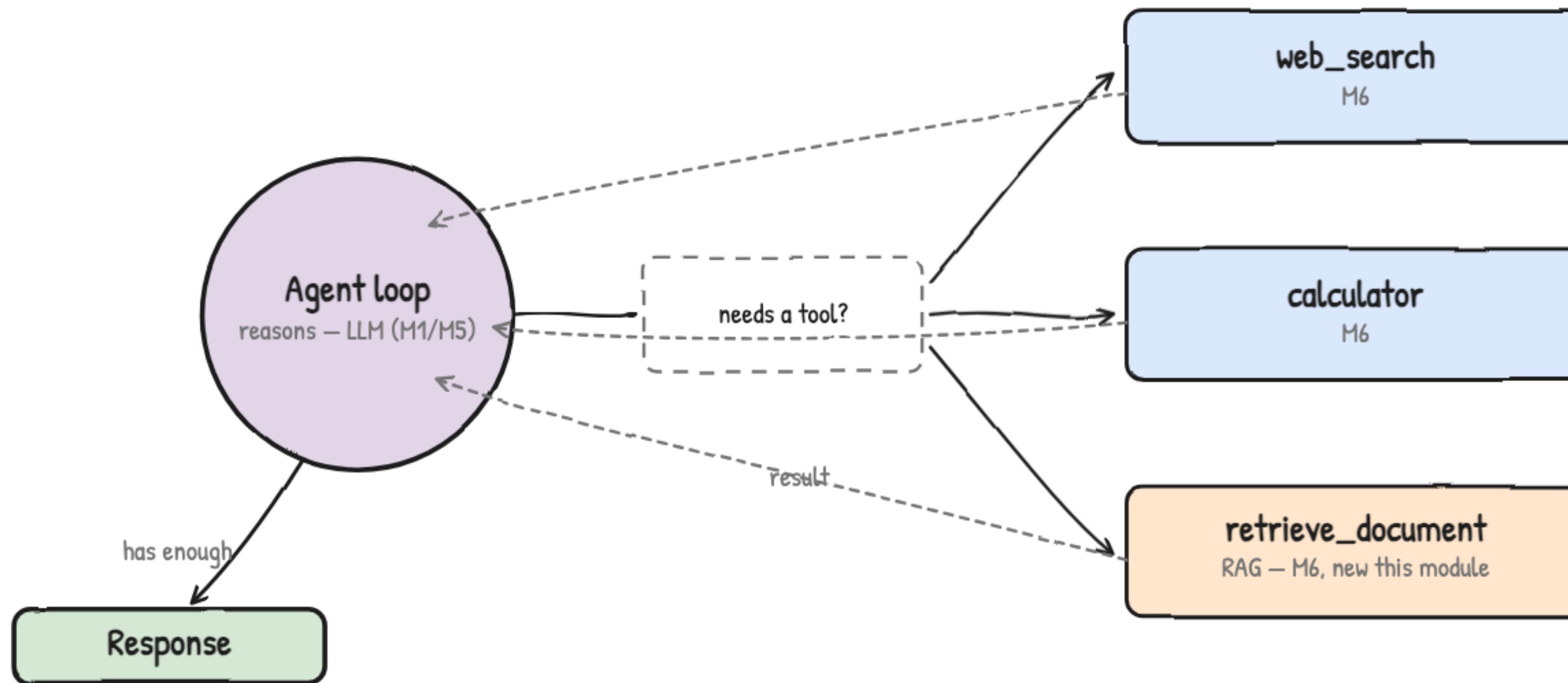
Read it left to right, one hop at a time.

Six modules converge behind one loop



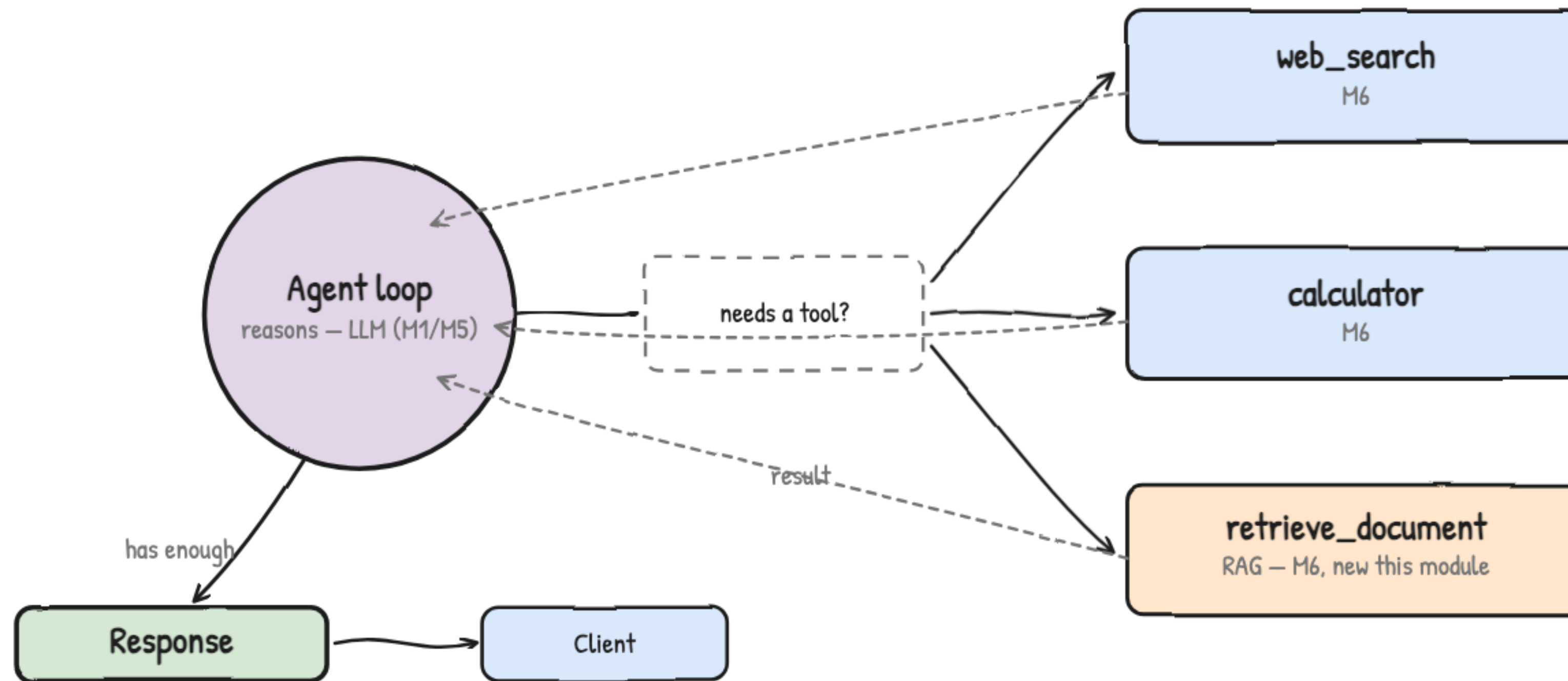
Read it left to right, one hop at a time.

Six modules converge behind one loop



Read it left to right, one hop at a time.

Six modules converge behind one loop



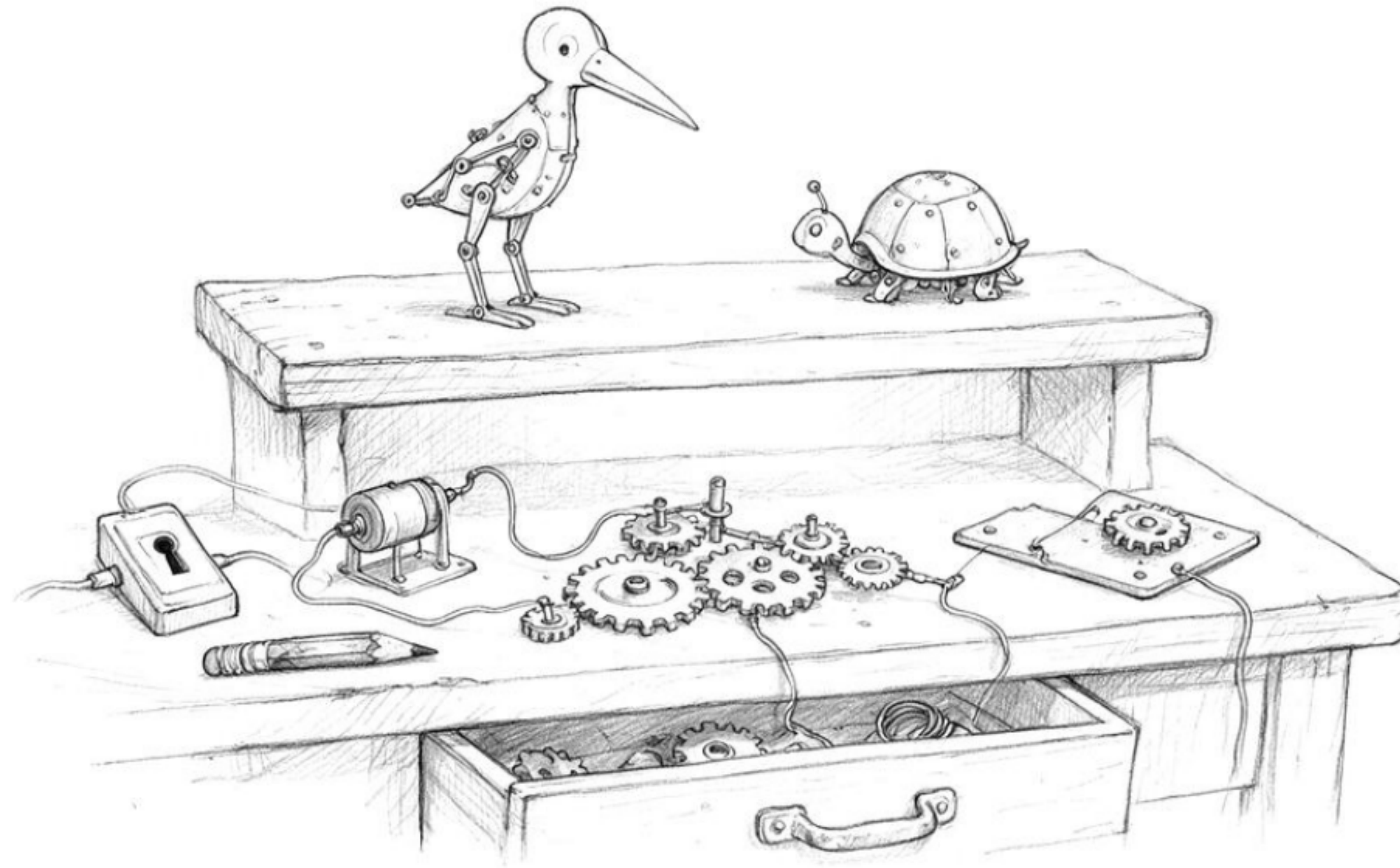
Read it left to right, one hop at a time.

Nothing here is new machinery — only new wiring

- 1 LLM call — M1, M5
- 2 agent loop — M6
- 3 three tools incl. RAG — M6
- 4 served behind an API — M7, M8

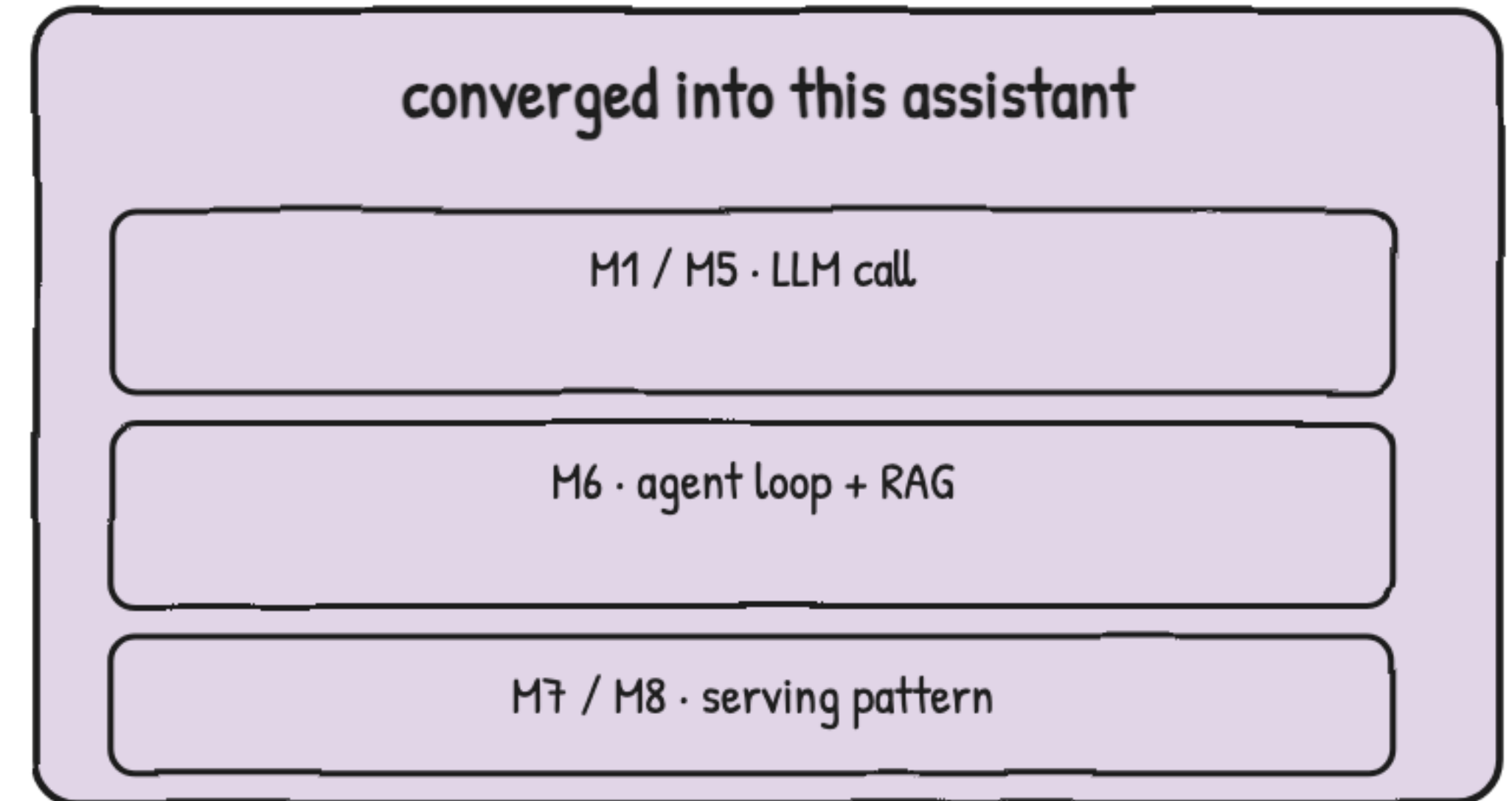
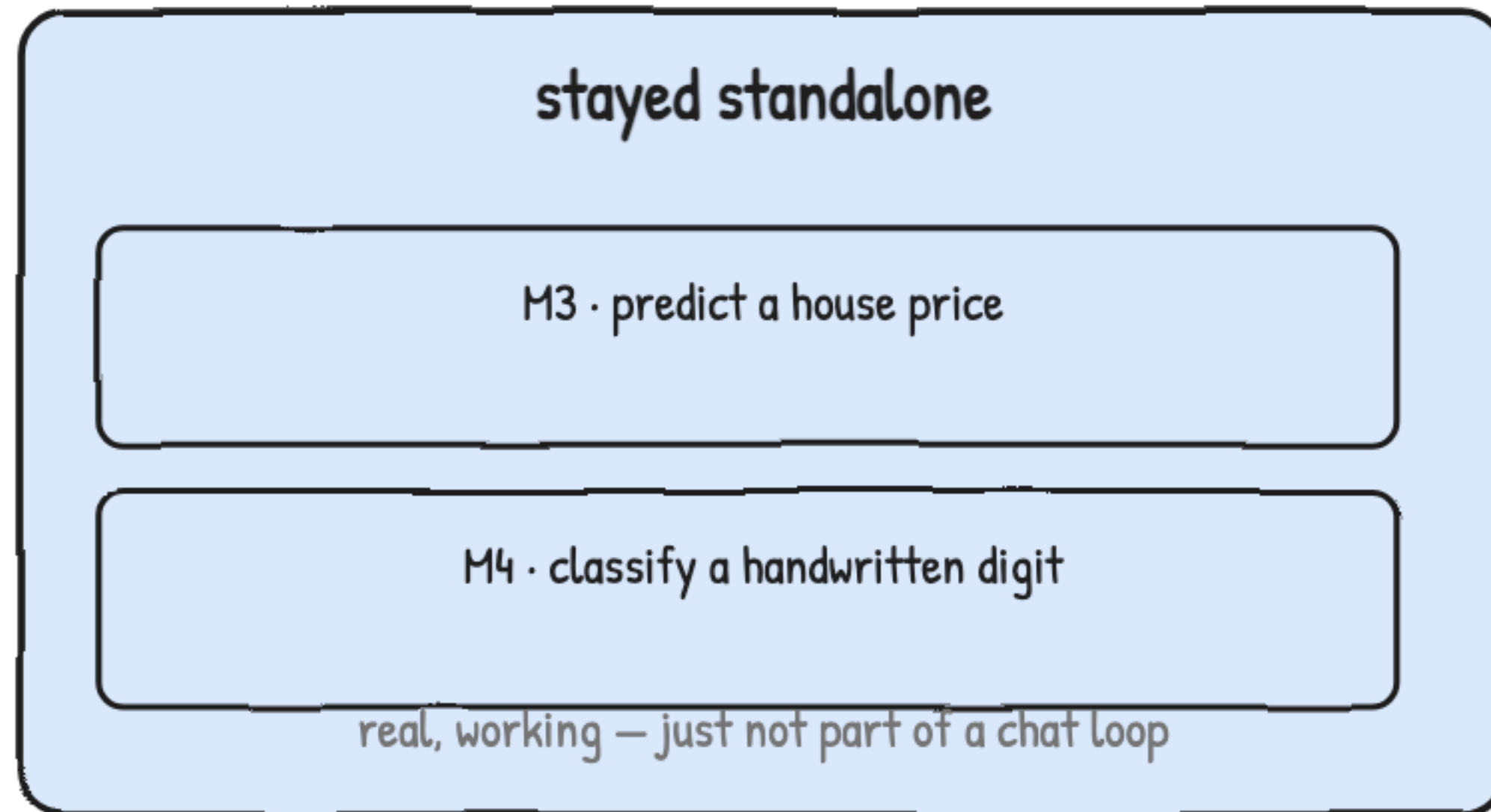
✓ four labs, one assistant

Every box traces to a lab you already ran.



The project spine, converged – and what didn't converge

What converged, and what stayed standalone



Both halves are real. Only one is a chat loop.



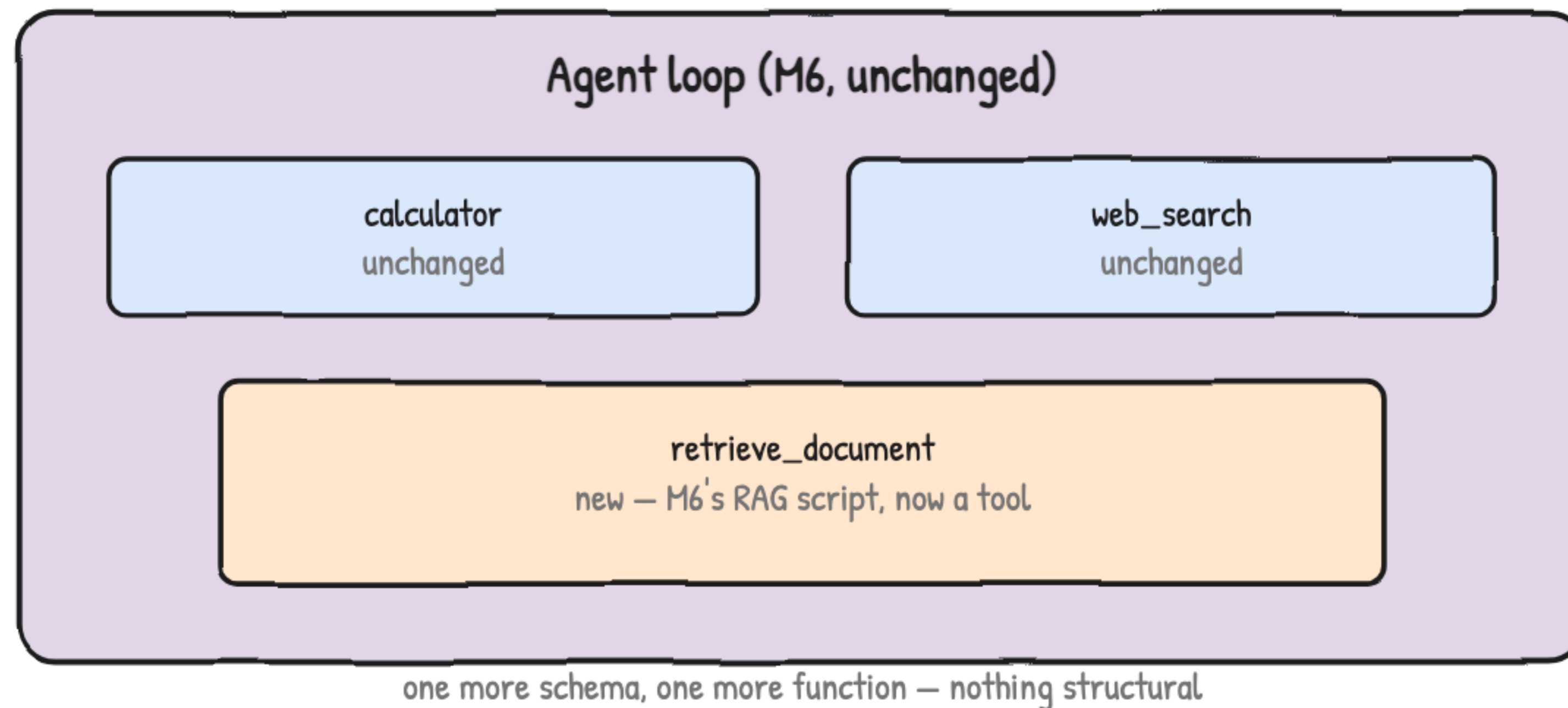
Three tools, one assistant

A good helpdesk doesn't answer everything the same way



It picks the right method for the question.

Three tools, three Python functions, one already-built loop

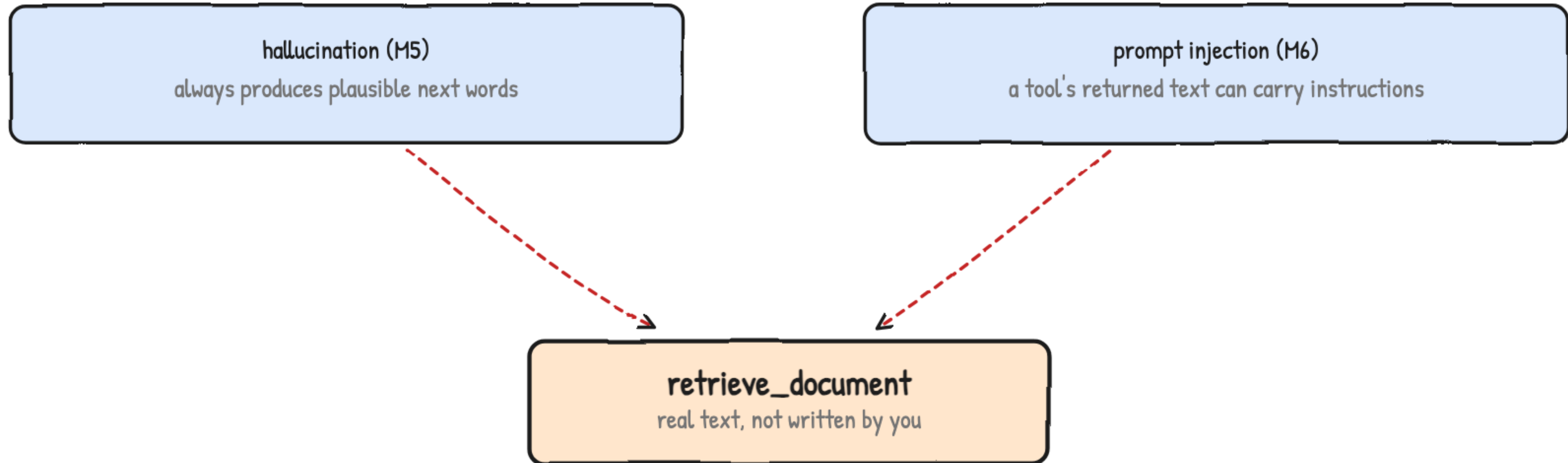


One more function. Nothing structural changes.



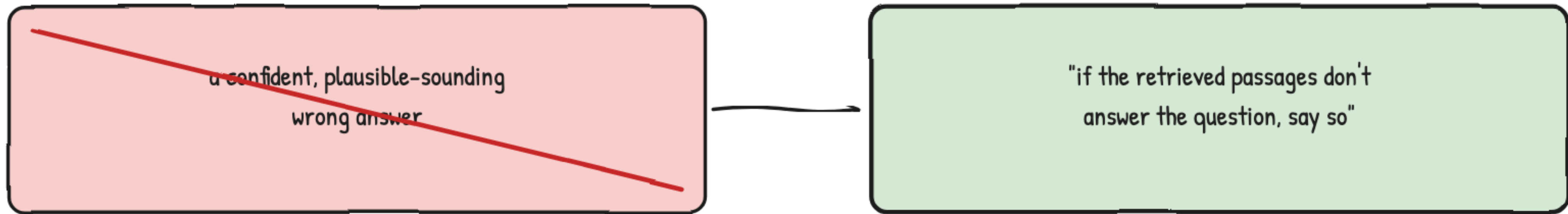
Honesty over confidence

Two risks meet at retrieve_document



Two risks you already know, one new tool.

The defense: a plain instruction, not a patch



the single most convincing thing you can show in a demo

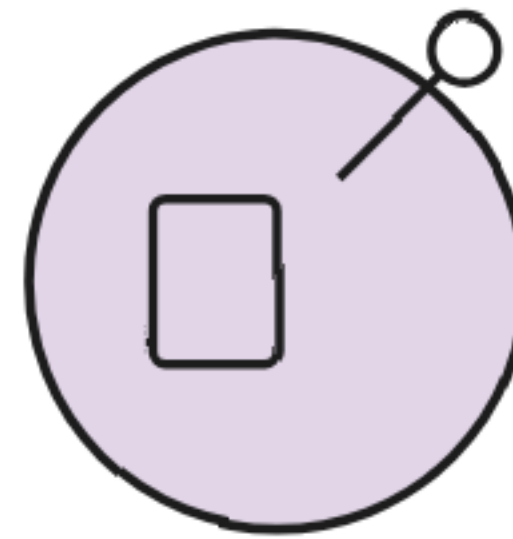
Say so, instead of a plausible guess.

A two-part build, then Demo Day



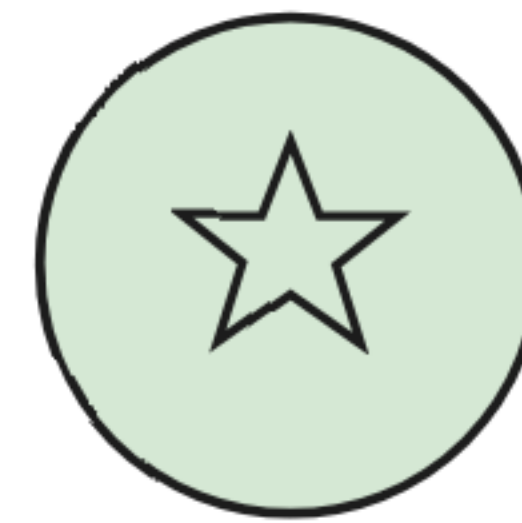
1 · Wire the Brain

LLM + agent loop



2 · Give It Knowledge & Hands

RAG + tools



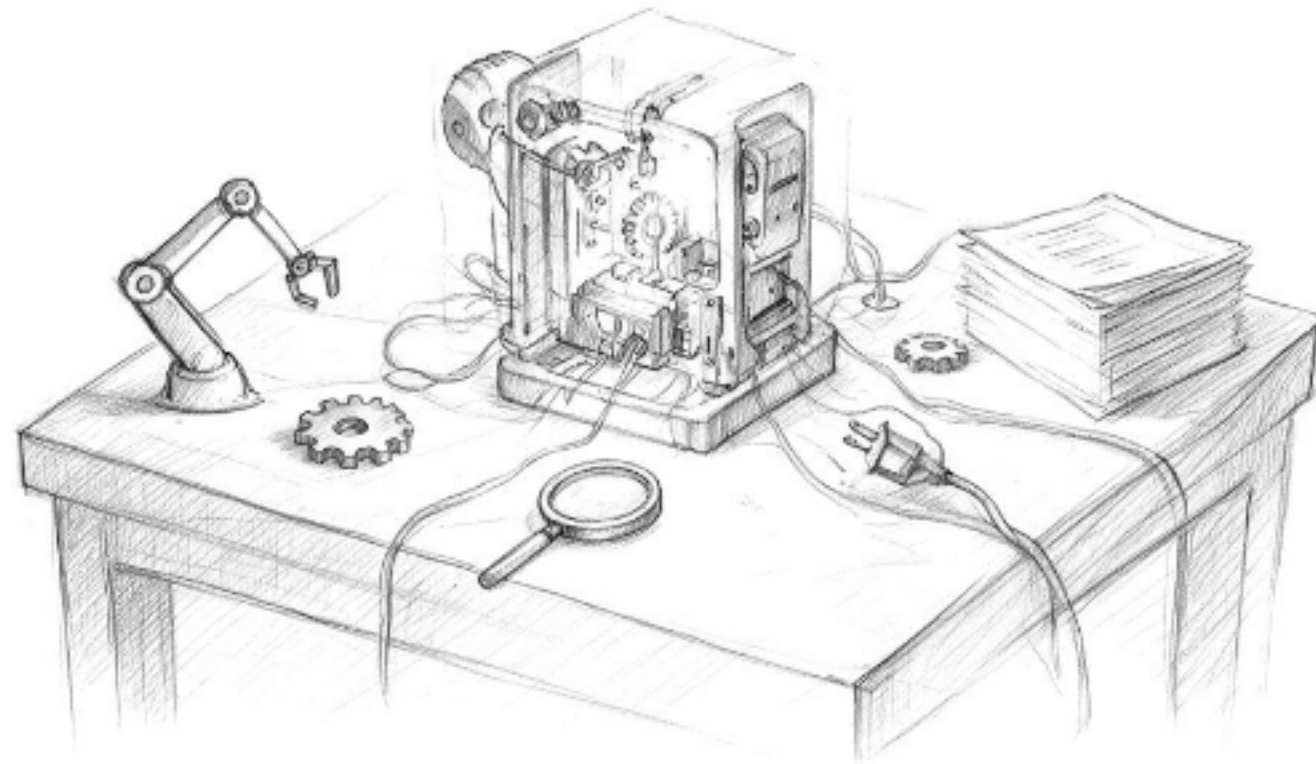
3 · Demo Day

run it, break it, show it off

Wire the brain, give it hands, then show it off.

MODULE 9 TAKEAWAYS

You now know exactly which lab built every piece of this assistant



- 1 LLM call — M1, M5
- 2 agent loop + RAG — M6
- 3 served behind an API — M7, M8
- 4 honesty over confidence

Now open Module 9 · Lab. · Gourav Shah · School of DevOps & AI